

# WASP-50b

R-1560

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_WASP-50\_141202 · created 2026-08-01T21:30:24+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

## Results

|                                                   |                               |
|---------------------------------------------------|-------------------------------|
| Mid-Transit Time (Tmid)                           | 2456993.741 +/- 0.073 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.168 +/- 0.05                |
| Transit depth (Rp/Rs)^2                           | 2.81 +/- 1.67 [%]             |
| Orbital Inclination (inc)                         | 89.64 +/- 2.91                |
| Airmass coefficient 1 (a1)                        | 0.62 +/- 0.43                 |
| Airmass coefficient 2 (a2)                        | 0.27 +/- 0.7                  |
| Scatter in the residuals of the lightcurve fit is | 62.6 %                        |
| Best Comparison Star                              | None                          |
| Optimal Aperture                                  | 1.21                          |
| Optimal Annulus                                   | 4.17                          |
| Transit Duration (day)                            | 0.039 +/- 0.036               |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                          |
|---------------------------|------------------------------------------|
| Rp/R■ fit vs published    | 0.168 ± 0.05 vs 0.1404 (deviation 0.55σ) |
| Duration fit vs published | 0.039 d vs 0.0754 d (deviation 1.01σ)    |
| Mid-time O–C              | 129.8 min                                |
| Depth SNR                 | 3.4                                      |
| Residual scatter          | 62.6 %                                   |

## Light curve

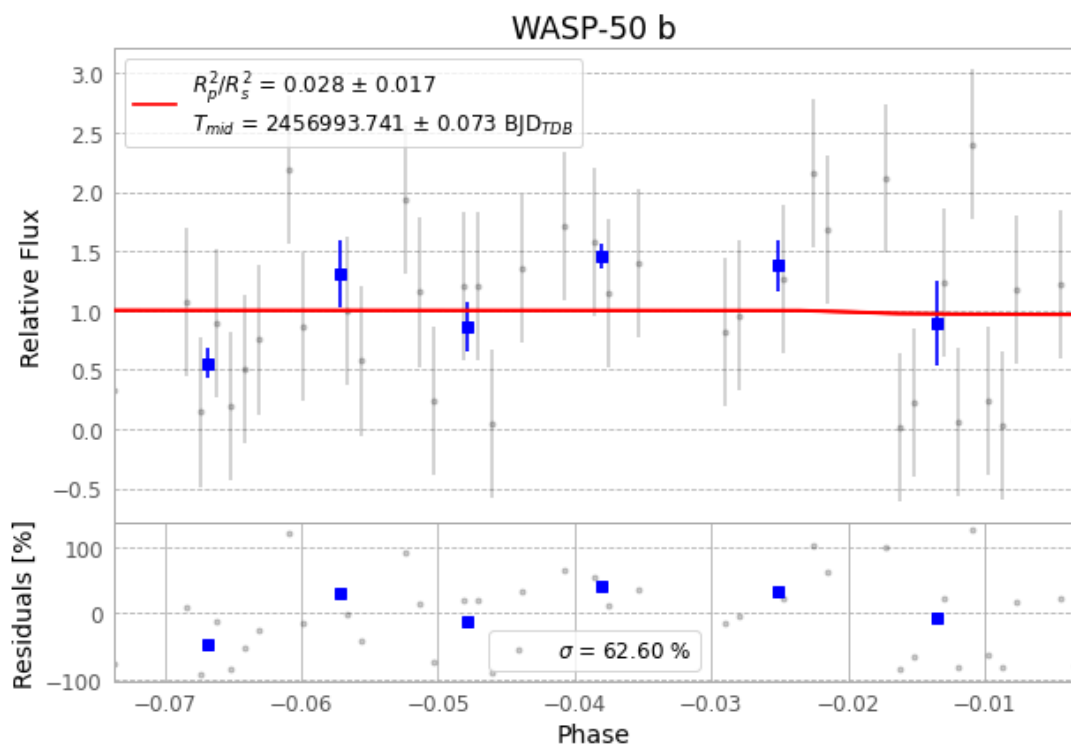


Figure 1 — FinalLightCurve\_WASP-50 b\_2014-12-01.png (SHA-256 81d014b6ee382657...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [554, 248], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 9643cd021e7adca1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ba3ef8b

## Data provenance

69 FITS frames (45 MB), WASP-50141202020912.FITS → WASP-50141202053312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-141202.log (c79736e96945...), FinalLightCurve\_WASP-50 b\_2014-12-01.png (81d014b6ee38...), FinalLightCurve\_WASP-50 b\_2014-12-01.csv (438fa61d5957...)

## Compute resources

Wall time 320.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 21:30Z.